

Use-Case Flow: Place Book Reservation/Hold

Use Case Name:

Place Book Reservation/Hold

Actors:

- Primary: Student Member
- Supporting: System (Notification Service), Campus SSO / Auth Server

Preconditions:

- Student Member is authenticated via Campus SSO.
- Member's outstanding fines do not exceed ₹500.
- The book (by ISBN) exists in the catalog and its status is “On Loan” or “Fully Reserved”.
- Member has fewer than 5 currently active reservations.

Postconditions:

- Success: A new reservation is created, appended to the FIFO queue at the next position, and status/queue position is visible to the member.
- Failure: No reservation is created; system state is unchanged; member sees a specific error message.

Main Success Scenario:

1. Student Member searches for a book by ISBN, title, or author and selects a title showing status “On Loan” or “Reserved.”
2. Student Member selects “Place Reservation/Hold” on the book’s detail page.
3. System verifies the member is authenticated via Campus SSO; if not, the member is redirected to log in and returns to this flow on success.
4. System checks the member’s account for outstanding fines (< ₹500) and active reservation count (< 5).
5. System checks that the member does not already hold an active reservation for the same ISBN.
6. System creates a reservation record with a unique reservation ID, timestamp, and queue position (current queue length + 1).
7. System updates the book’s availability display to reflect the member’s queue position.
8. System displays a confirmation screen showing reservation ID, book title, and queue position.
9. System logs the transaction for audit purposes.
10. Use case ends successfully.

Alternate Flow:

4a. Maximum Reservation Limit Reached

Trigger: At Step 4, the system detects the member already has 5 active reservations.

4a1. System halts the reservation process; no reservation record is created.

4a2. System displays: “You have reached the maximum limit of 5 active reservations. Please cancel an existing reservation or wait for a hold to be fulfilled.”

4a3. System presents the member’s active reservations with an option to cancel one.

4a4. If the member cancels a reservation, the flow returns to Step 2 for the original book.

4a5. If the member does not cancel, the use case ends (failure postcondition).

Related Requirements: FR-002, FR-003, FR-004, NFR-002